

Sina Sajadmanesh

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SUMMARY

Senior AI Engineer specializing in multimodal foundation models, with a strong focus on translating research into high-performance, production-ready systems. Proven track record in building scalable model training pipelines and optimizing inference across heterogeneous hardware.

EDUCATION

Swiss Federal Institute of Technology (EPFL), PhD in Electrical Engineering Aug 2023

Sharif University of Technology, MSc in Information Technology Engineering Sept 2016

University of Isfahan, BSc in Computer Software Engineering Feb 2014

EXPERIENCE

Sony AI, Senior AI Engineer – Zurich, Switzerland Jan 2026 – present

- Built agentic AI workflows for automated model optimization, improving inference speed and memory footprint of LLMs and VLMs for Apple Silicon
- Architected a modular ML acceleration and deployment framework supporting diverse hardware targets (Nvidia GPUs, Qualcomm NPUs, Apple Silicon)
- Developed compact vision-language models supporting image/video understanding and localization tasks, achieving competitive performance with reduced computational requirements

Sony AI, AI Engineer – Zurich, Switzerland Oct 2023 – Dec 2025

- Designed a modular and flexible geometric optimization algorithm, enabling NeurIPS'25 spotlight paper
- Led end-to-end optimization of Sony's Argus VFM inference stack, resulting in up to $61\times$ speedup across 17 vision tasks
- Developed 7 computer vision models, including object detection, depth estimation, super-resolution, and OCR, achieving SOTA or competitive results
- Architected and implemented a foundational MTL framework for Sony's Argus VFM, supporting 17 vision tasks, enabling CVPR'25 publication

Idiap Research Institute, Research Assistant – Martigny, Switzerland May 2019 – Aug 2023

- Worked on differentially private machine learning with graph neural networks
- 3 papers in top-tier conferences (CCS, USENIX Security, and WSDM) with 340+ citations
- 7 invited talks at top universities and research institutions, including Imperial College, UIC, and Twitter
- 6 open-source projects with 130+ stars on GitHub
- 1 short course taught on "Trustworthy Machine Learning" at Artificial Intelligence Doctoral Academy
- Finalist in CSAW Applied Research Competition for the best paper award in computer security in Europe

Brave Software, Research Intern – Remote Mar 2022 – May 2022

- Developed a novel privacy-preserving federated learning framework for neural bandit models under client heterogeneity using PyTorch, Flower, Dask, and Tensorflow-Lite

SKILLS AND EXPERTISE

Foundation & Generative Models: Vision & Multimodal Models (ViT, CLIP, LLaVA) · Large Language Models (LLaMA, GPT, Qwen) · Parameter-Efficient Fine-Tuning (LoRA) · Self-Supervised Learning (MAE, DINO)

Model Optimization & Deployment: TensorRT · ONNX · Qualcomm QNN / SNPE / QAIIR · Apple CoreML / MLX · Optimum · Quantization · Graph Optimization · Performance Profiling

Machine Learning Frameworks: PyTorch · Hugging Face (Transformers, PEFT, Accelerate, Datasets) · OpenMMLab (MMCV, MMEEngine, MMDetection)

Agentic AI & LLM Systems: LangChain · LangGraph · MCP · Tool Calling · RAG · Multi-Agent Systems · Workflow Orchestration

Engineering & Tooling: Python · C++ · Bash · SQL · Data Processing (NumPy, Pandas, torchvision) · Environment & Packaging (uv, ruff) · CI/CD (Docker, GitHub Actions)

COMMUNITY AND PROFESSIONAL SERVICE

Invited Speaker: Imperial College London (2023, 2020), University of Illinois at Chicago (2022), L3S Research Center (2022), GNN User Group Meetup (2021), Twitter ML Seminar (2021)

Organizing / Program Committee: Privacy and Fairness in AI for Health (2023), PPAI (2024), WiseML (2023), PAIR2Struct (2022), DPML (2021)

Reviewer: NeurIPS (2026, 2024), CVPR (2026), TIFS (2025), IMWUT (2024), TDSC (2023), LoG (2023, 2022), AISTATS (2023), AIJ (2022), TBD (2021), TIST (2020), SNAM (2020), WWWJ (2018)

SELECTED PUBLICATIONS

Stella: Subspace Learning in Low-rank Adaptation using Stiefel Manifold Dec 2025
Zhizhong Li, Sina Sajadmanesh, Jintao Li, Lingjuan Lyu
NeurIPS 2025 🌟 Spotlight (top 3%)

Argus: A Compact and Versatile Foundation Model for Vision June 2025
Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, et al.
CVPR 2025

ProGAP: Progressive Graph Neural Networks with Differential Privacy Guarantees Mar 2024
Sina Sajadmanesh, Daniel Gatica-Perez
WSDM 2024

GAP: Differentially Private Graph Neural Networks with Aggregation Perturbation Aug 2023
Sina Sajadmanesh, Ali Shahin Shamsabadi, Aurélien Bellet, Daniel Gatica-Perez
USENIX Security 2023

Locally Private Graph Neural Networks Nov 2021
Sina Sajadmanesh, Daniel Gatica-Perez
CCS 2021

A Hybrid Deep Learning Architecture for Privacy-Preserving Mobile Analytics May 2020
Seyed Ali Osia, Ali Shahin Shamsabadi, Sina Sajadmanesh, et al.
IEEE Internet of Things Journal

Continuous-Time Relationship Prediction in Dynamic Heterogeneous Information Networks Aug 2019
Sina Sajadmanesh, Sogol Bazargani, Jiawei Zhang, Hamid R. Rabiee
TKDD

Kissing Cuisines: Exploring Worldwide Culinary Habits on the Web Apr 2017
Sina Sajadmanesh, Sina Jafarzadeh, Seyed Ali Ossia, et al.
WWW 2017